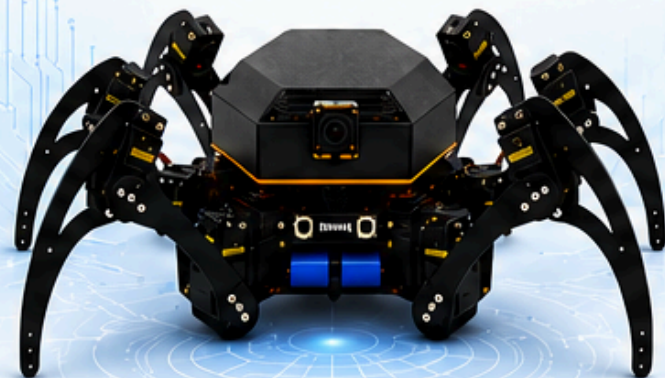




HEXAGLAM_K

SMART HEXAPOD ROBOT

SIX LEGS. SMART SENSES. ENDLESS POSSIBILITIES.



ABOUT THE PROJECT

HEXAGLAM_K is a smart, six-legged robot integrated with multiple sensors, live camera streaming and mobile control using MIT App Inventor. It is designed for real-time monitoring, obstacle detection and data visualization with a stable and adaptable walking mechanism for various terrains.

KEY FEATURES

- Bluetooth & Wi-Fi Control
- Live Camera Streaming (ESP32-CAM)
- Multiple Sensor Monitoring
- Real-time Data Display
- 6-Legged Stable Movement
- MIT App Inventor Mobile Application
- ESP32 Based Controller
- Obstacle Detection & Navigation
- Battery Monitoring
- Expandable Architecture



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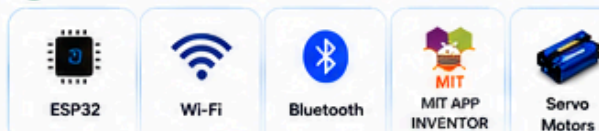
INTRODUCTION

HexaGLAM_K is a multi-functional hexapod robot built on ESP32 microcontroller with advanced sensing capabilities and real-time monitoring. It is capable of navigating rough terrains, detecting obstacles, and streaming live video to the user.

OBJECTIVES

- Develop a stable and efficient six-legged walking robot.
- Integrate multiple sensors for environmental monitoring.
- Enable real-time video streaming and data display.
- Provide mobile control using MIT App Inventor.

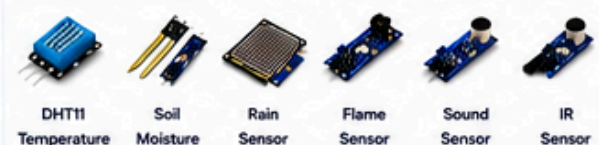
TECHNOLOGIES USED



HARDWARE COMPONENTS



SENSORS USED



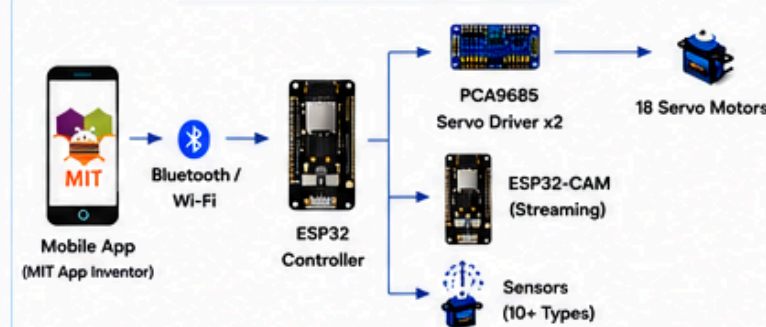
SOFTWARE USED



TECHNICAL SPECIFICATIONS

Item	Specification	Item	Specification
Controller	ESP32	Mobile App	MIT App Inventor
Camera	ESP32-CAM OV3660	Battery	7.4V Li-Ion / 2200mAh (Typical)
Servo Motors	18 x SG90	Programming	Arduino C++
Servo Driver	PCA9685 x 2	Walking Modes	Manual & Autonomous
Communication	Bluetooth + Wi-Fi	Dimensions	(Approx.) 32cm x 32cm x 12cm
		Weight	(Approx.) 1.2 kg

SYSTEM ARCHITECTURE



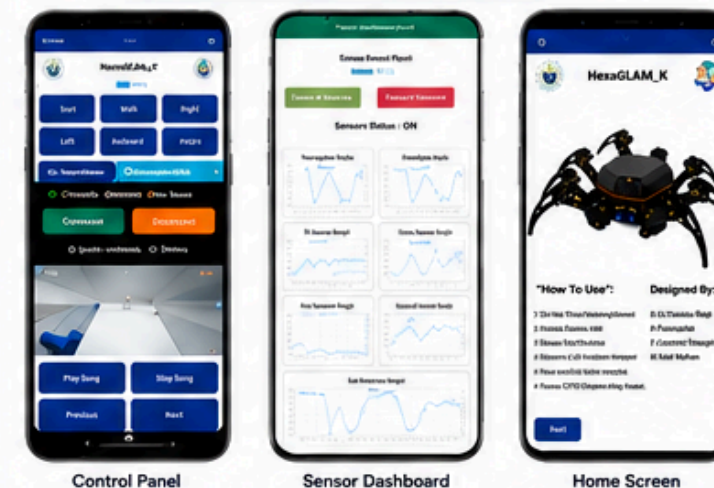
HOW IT WORKS (STEP-BY-STEP)

- User opens the MIT App.
- Connects to ESP32 via Bluetooth / Wi-Fi.
- Sends movement or action command.
- ESP32 processes the command.
- PCA9685 controls servo motors.
- Hexapod moves on the desired direction.
- Sensors continuously monitor environment.
- Camera streams live video to the app.
- Data & status displayed in real-time.

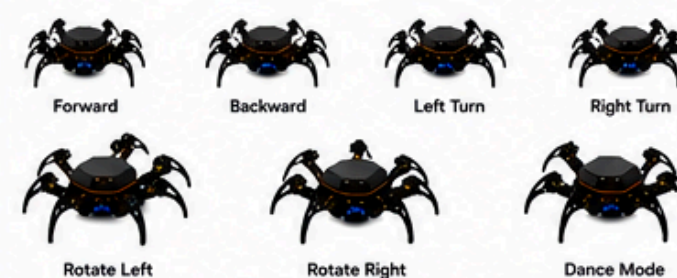
SYSTEM WORKFLOW



MIT APP INTERFACE OVERVIEW



ROBOT MOVEMENT FEATURES



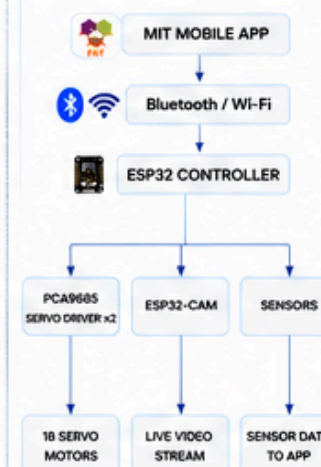
ADVANTAGES

- Stable six-legged locomotion
- Multiple integrated sensors
- Remote mobile control
- Live camera monitoring
- Real-time environmental data
- Expandable & modular design
- Energy efficient & reliable
- Suitable for rough terrains

APPLICATIONS

- Military Surveillance
- Search & Rescue Operations
- Industrial Inspection
- Smart Agriculture
- Disaster Management
- Security Patrol
- Research & Education

COMMUNICATION FLOW



FUTURE SCOPE

- AI-based object detection
- Autonomous navigation
- GPS tracking & mapping
- Voice command control
- Cloud data storage & analytics
- Robotic arm integration

TEAM DETAILS



Dr. K. Krishna Naik



Team Members

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Project Video